

No-NN Ablation Study – Correction Note

1. Correction to NN Invocation Fraction

In the original summary table, the column labeled 'NN_frac' for the No-NN baseline was incorrectly reported as ~0.0047. This value actually represents the fraction of close pair interactions ($r <$ threshold), not neural network usage. For the No-NN baseline, the NN is never invoked, so the correct NN invocation fraction is 0.0000. To improve clarity, the metrics are now separated as follows:

- NN_frac: fraction of pair evaluations where the NN is used
- close_pair_frac: fraction of pair evaluations within the close-encounter region

2. Updated Interpretation

The corrected results confirm that both SIMON and the No-NN baseline experience similar close-encounter frequencies, but only SIMON uses the NN during those steps. This reinforces that the NN is a targeted correction mechanism rather than a dominant component.

3. Minor Wording Fix

The phrase 'both bodies remain bounded' has been corrected to 'all bodies remain bounded', since the system contains three bodies.

4. Final Scientific Conclusion (Unchanged)

Adaptive sub-stepping remains the primary mechanism ensuring orbital stability. The neural network provides a localized accuracy improvement during close encounters, reducing final RMS error without affecting the divergence rate.